

RUDRESH GAWAS

rudreshgawas3@gmail.com | +91 7066158254 | Sanquelim, Goa | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Information Technology engineering student (Graduation: June 2026, CGPA: 7.5/10) with hands-on industry experience at Siemens and a strong project track record spanning AI, IoT, and full-stack development. Led a team of 4 to engineer PhysioCare — a cross-platform AI physiotherapy rehabilitation platform using Flutter, MediaPipe, and Google ML Kit. Built agentic AI systems using CrewAI, LangGraph, AutoGen, and PydanticAI — including multi-agent pipelines, RAG systems, and ReAct agents. Proficient in Python, FastAPI, Flutter, NLP (HuggingFace Transformers), and IoT systems. Demonstrated ability to deliver end-to-end solutions — from real-time pose estimation to data pipelines, automated reporting, and AI-driven classification. Seeking a full-time role in, AI-Full-Stack Developer, AI/ML Engineering.

EXPERIENCE

Software Developer Intern | Siemens

Goa, India | July 2025 – August 2025

Real-Time Production Monitoring Dashboard

- Designed and deployed a real-time dashboard integrating live server log data from 3 machine types (AOI SAKI, MEK, SPI SAKI), reducing manual monitoring effort by 70% across the production line.
- Built a data pipeline that processed thousands of machine-generated log files daily, surfacing live production status, machine health, and product insights for floor managers.
- Tech stack: Python, Streamlit, SQLite, Pandas.

Component Lookup Tool

- Developed a web-based internal tool enabling manufacturing teams to instantly retrieve component details via product IDs, eliminating manual lookup and reducing errors on time-sensitive components (23-hour post-exposure window).
- Tech stack: JavaScript, HTML, CSS.

PROJECTS

PhysioCare — AI-Powered Physiotherapy Rehabilitation Platform — Flutter, Google ML Kit, MediaPipe

BlazePose, Supabase, Gemini 2.5 Flash, Dart PDF, flutter_tts

2025–2026

- Led a team of 4 as Team Lead to architect and deliver a cross-platform Flutter rehabilitation app — engineered end-to-end from database schema design to UI layer, driving sprint planning, task delegation, and code reviews throughout the project lifecycle.
- Engineered a dual pose estimation pipeline using Google ML Kit (Android/iOS) and MediaPipe BlazePose (Web) — detecting 33 skeletal landmarks per frame at sub-second latency to evaluate exercise form in real time.
- Developed angle-based rep-counting logic with state machines and cooldown debouncing for 5 exercise types: Bicep Curl, Side Raise, Squats, Standing Hip Abduction, and Seated Knee Extension.
- Integrated live accuracy scoring engine, AI clinical progress notes via Google Gemini 2.5 Flash, downloadable PDF rehabilitation reports, cross-platform Text-to-Speech audio guidance, and a real-time pain alert system notifying therapists and family members.
- Built a full therapist clinical dashboard enabling exercise plan assignment, session report review, patient feedback, and pain alert monitoring — with all session data persisted to Supabase (PostgreSQL).

ComplaintIQ — AI-Powered Banking Complaint Management System — Python, FastAPI, Streamlit,

HuggingFace Transformers, PyTorch, SQLite, Plotly, FPDF, bcrypt

2025

- Built a full-stack AI complaint management system for the banking sector that automates ticket classification, prioritization, and resolution using NLP.
- Integrated a PyTorch-based HuggingFace Transformers sentiment analysis pipeline to automatically detect customer sentiment (Angry / Neutral / Happy) and route tickets to the correct department (Loan, Credit Card, Fraud, Account) — reducing manual triage time.
- Implemented urgency detection logic to instantly flag high-priority cases (e.g., fraud), ensuring critical tickets are escalated without human intervention.
- Built a batch processing engine using Pandas to handle CSV uploads and process thousands of complaints simultaneously in seconds.

- Auto-generated professional PDF Executive Summary reports using FPDF for management review, and built an interactive analytics dashboard with Plotly for real-time case metrics.
- Secured the system with bcrypt password hashing, role-based access control (Agent login/registration), and environment variable isolation to prevent credential leaks.

IoT AccessGuard — Smart Attendance & Door Security System — ESP32, ESP8266, ESP32-CAM, RFID, Arduino IDE, Google Sheets API, Blynk IoT, ML 2024–2025

- Engineered a unified IoT product integrating 5 authentication methods — RFID, keypad, fingerprint sensor, face recognition (ML), and IoT mobile app — into a single compact door lock and attendance system.
- Implemented real-time attendance logging via Google Sheets API over Wi-Fi, eliminating manual record-keeping.
- Selected for DIPEX 2025 State-Level Exhibition & Competition, Pune after clearing the Ratnagiri district round — presented to industry and academic judges.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, SQL, HTML5, CSS3, PHP

AI / NLP / ML: HuggingFace Transformers, PyTorch, CrewAI, LangChain, LangGraph, AutoGen (AG2), PydanticAI, Gradio, Docling, Sentiment Analysis, Custom Dataset Training, RoboFlow, OpenCV

Backend & APIs: FastAPI, Uvicorn, Node.js, REST APIs, SMTP, Python-multipart

Frontend: Streamlit, Flutter, React.js, HTML5, CSS3, Plotly

Data & Databases: Pandas, SQLite, MySQL, Supabase, Google Sheets API, FPDF, SciPy, CSV/JSON

IoT & Hardware: ESP32, ESP8266, ESP32-CAM, RFID, Arduino IDE, Blynk IoT

Security: bcrypt, Role-Based Access Control, Environment Variable Management (.env)

Tools & Platforms: Git, GitHub, VS Code, Google Colab, Postman, npm, Streamlit CLI

EDUCATION

Bachelor of Engineering — Information Technology

2022 – June 2026 (Expected)

Shree Rayeshwar Institute of Engineering and Information Technology, Shiroda, Goa | **CGPA: 7.5 / 10**

ACHIEVEMENTS & COMPETITIONS

- DIPEX 2025 State-Level Exhibition & Competition, Pune — IoT AccessGuard selected after clearing Ratnagiri district round (organized by ABVP & Srijan Trust).
- Science Film Festival of India (SCIFF 2025) — presented two independent IoT engineering projects to national-level panel.
- CONVERGE 2.0 2025 — participant, organized by Directorate of Higher Education, Goa State Higher Education Council.

CERTIFICATIONS

- **Tata Group Data Analytics Job Simulation** — Forage (Jan 2026)
- **IBM SkillsBuild-AI Strategy & Business Intelligence Internship** — AICTE|CSRBOX (April 2026)
- **Agentic AI Hands-on** — IBM Skills Network (1 course, 9 guided projects) (May 2026)
- **Applied Artificial Intelligence — AI Agent Development** — CSRBOX Winter Program (2025)
- **Data Science Master Virtual Internship** — AICTE-EDUSKILLS (2024)
- **AI-ML Virtual Internship** — AICTE-EDUSKILLS (2024)
- **Problem Solving through Programming in C** — NPTEL (2023)
- **The Joy of Computing Using Python** — NPTEL (2023)